

Static Gesture Classification — Run Report

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Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Static · 35 trials
Training data path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Static · 35 trials
Training data path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Static · 35 trials
Training data path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Static · 35 trials
Training data path 5	/home/jestin/ThesisRepo/ML/NewTestData/5_Georgia/Static · 35 trials
Training data path 6	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Static · 35 trials
Training data path 7	/home/jestin/ThesisRepo/ML/NewTestData/7_Henry/Static · 35 trials
Training data path 8	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Static · 36 trials
Training data path 9	/home/jestin/ThesisRepo/ML/NewTestData/9_Joselyn/Static · 30 trials
Training data path 10	/home/jestin/ThesisRepo/ML/NewTestData/10_Josh/Static · 35 trials
Training data path 11	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Static · 36 trials
Training data path 12	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Static · 35 trials
Training data path 13	/home/jestin/ThesisRepo/ML/NewTestData/13_Nat/Static · 37 trials
Training data path 14	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Static · 35 trials
Training data path 15	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Static · 0 trials
Training data path 16	/home/jestin/ThesisRepo/ML/NewTestData/16_Bella/Static · 35 trials
Training data path 17	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Static · 35 trials
Total training paths	17 · 559 trials total
External test root	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Static
Report output dir	/home/jestin/ThesisRepo/ML/NewReports/StaticReports

Sensor selection

Hands	left, right
Segments	palm_mid, thumb, index, middle, ring, pinky, wrist
Modalities	YPR, Accel, Flex
Sensor columns	176 channels
Classes	7: Double_Closed_Fist, Double_Nothing, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman

Preprocessing, features & split

Resample steps	90
Butterworth	off · cutoff 10.0 Hz · order 4 · fs 30.0 Hz
Normalisation	minmax
Feature mode	stats+fft → feature dim 9504
Test size · seed · CV folds	0.2 · random_state=42 · 10-fold stratified
Sample counts	train (post-aug)=2235 · test=112

Augmentation (training only)

Enabled	yes
Copies per train trial	4
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.9, 1.1)
Time warp	off · range=(0.9, 1.1)

Classifiers — cross-validation & hold-out test

Algorithm	CV mean (10-fold)	CV std	Hold-out test acc
Random Forest	1.0000	0.0000	0.9821
SVM (Linear)	1.0000	0.0000	0.9732
SVM (RBF)	1.0000	0.0000	0.9554
Logistic Regression	1.0000	0.0000	0.9464
LDA	1.0000	0.0000	0.7768
KNN	0.9911	0.0080	0.7679

Per-class hold-out F1

Class	Random Forest	SVM (Linear)	SVM (RBF)	Logistic Regression	LDA	KNN	Support
Double_Closed_Fist	1.000	0.971	0.941	0.919	0.857	0.778	17
Double_Nothing	0.929	0.889	0.897	0.889	0.500	0.421	15
Double_Okay	1.000	0.970	0.941	0.970	0.706	0.625	16
Double_Open_Palm	1.000	1.000	0.968	0.968	0.684	0.778	16
Double_Phone	1.000	1.000	0.968	0.933	0.824	0.774	16
Double_Pistol	0.941	0.970	0.970	0.970	0.933	0.865	16
Double_Spiderman	1.000	1.000	1.000	0.970	0.848	0.970	16

macro avg	0.981	0.971	0.955	0.945	0.765	0.744	112
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External test (NewTestData)

Algorithm	External test acc
SVM (Linear)	1.0000
Random Forest	1.0000
LDA	0.9714
SVM (RBF)	0.9429
KNN	0.9143
Logistic Regression	0.9143

Per-class external accuracy

Class	SVM (Linear)	Random Forest	LDA	SVM (RBF)	KNN	Logistic Regression	Support
Double_Closed_Fist	1.000	1.000	1.000	1.000	1.000	1.000	5
Double_Nothing	1.000	1.000	0.800	1.000	0.800	0.800	5
Double_Okay	1.000	1.000	1.000	1.000	1.000	1.000	5
Double_Open_Palm	1.000	1.000	1.000	0.800	0.800	0.800	5
Double_Phone	1.000	1.000	1.000	0.800	0.800	0.800	5
Double_Pistol	1.000	1.000	1.000	1.000	1.000	1.000	5
Double_Spiderman	1.000	1.000	1.000	1.000	1.000	1.000	5